

```
In [1]: import pandas as pd

df = pd.read_csv(
    "bank_marketing/bank-full.csv",
    sep=';'
)

print(df.head())
print(df.shape)
```

Out:

	age	job	marital	education	default	balance	housing	loan	\
0	58	management	married	tertiary	no	2143	yes	no	
1	44	technician	single	secondary	no	29	yes	no	
2	33	entrepreneur	married	secondary	no	2	yes	yes	
3	47	blue-collar	married	unknown	no	1506	yes	no	
4	33	unknown	single	unknown	no	1	no	no	

	contact	day	month	duration	campaign	pdays	previous	poutcome	y
0	unknown	5	may	261	1	-1	0	unknown	no
1	unknown	5	may	151	1	-1	0	unknown	no
2	unknown	5	may	76	1	-1	0	unknown	no
3	unknown	5	may	92	1	-1	0	unknown	no
4	unknown	5	may	198	1	-1	0	unknown	no

(45211, 17)

```
In [2]: from sklearn.preprocessing import LabelEncoder
```

```
le = LabelEncoder()

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col])

print(df.head())
```

Out:

	age	job	marital	education	default	balance	housing	loan	contact	\
0	58	4	1	2	0	2143	1	0	2	
1	44	9	2	1	0	29	1	0	2	

2	33	2	1	1	0	2	1	1	2
3	47	1	1	3	0	1506	1	0	2
4	33	11	2	3	0	1	0	0	2

	day	month	duration	campaign	pdays	previous	poutcome	y
0	5	8	261	1	-1	0	3	0
1	5	8	151	1	-1	0	3	0
2	5	8	76	1	-1	0	3	0
3	5	8	92	1	-1	0	3	0
4	5	8	198	1	-1	0	3	0

```
In [3]: X = df.drop('y', axis=1)
        y = df['y']
```

```
In [4]: from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

```
In [5]: from sklearn.tree import DecisionTreeClassifier
```

```
model = DecisionTreeClassifier(random_state=42)
```

```
model.fit(X_train, y_train)
```

Out: ▼ DecisionTreeClassifier ⓘ ?

► Parameters

```
In [6]: y_pred = model.predict(X_test)
```

```
In [7]: from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(y_test, y_pred)
```

Out: Accuracy: 0.8740462235983634

```
plt.figure(figsize=(20,10))
```

```
plt.savefig("decision_tree.png")
plt.show()
```

Out:

